

Material Science and

□ To impart the basic concepts of metallurgy and materials of construction in chemical

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5073C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5073C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	5073C
Course title	Material Science and
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the basic concepts of metallurgy and materials of construction in chemical

Course outcomes

CO	Official outcome	Study level
CO1	Summarize the structural properties of crystals 13 Understanding	See official syllabus
CO2	Summarize the properties of materials 17 Understanding Summarize the processing and application of metal	See official syllabus
CO3	17 Understanding alloys Summarize the characteristics and processing of	See official syllabus
CO4	13 inorganic materials Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarize the structural properties of crystals	4	As specified
Module 2	Summarize the properties of materials	4	As specified
Module 3	Summarize the processing and application of metal alloys	4	As specified
Module 4	Summarize the characteristics and processing of inorganic materials	4	As specified

MODULE 1

Summarize the structural properties of crystals

This module is presented from the official syllabus scope for Material Science and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the structural properties of crystals
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding technology.

2 Understanding technology. is an official topic in Module 1 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding technology. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding technology. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding technology. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding technology.
 definition/meaning . steps, application . Technical terms English-
 .

Identify the various materials and its applications 3 Understanding.

Identify the various materials and its applications 3 Understanding. is an official topic in Module 1 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the various materials and its applications 3 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the various materials and its applications 3 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the various materials and its applications 3 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- Identify the various materials and its applications 3 Understanding. definition/meaning . steps, application . Technical terms English-

Outline crystal structure 4 Understanding.

Outline crystal structure 4 Understanding. is an official topic in Module 1 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline crystal structure 4 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline crystal structure 4 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline crystal structure 4 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓൗ Outline crystal structure 4 Understanding.

ഓൗൗൗൗൗൗൗൗൗ ഓൗൗൗ definition/meaning ഓൗൗൗൗൗൗൗൗൗ. ഓൗൗൗൗ ഓൗൗൗൗൗ ഓൗൗൗൗൗൗ, steps, application ഓൗൗൗൗ ഓൗൗൗൗൗൗൗൗൗൗൗ. Technical terms English-ഓൗ ഓൗൗൗൗൗൗൗൗൗൗൗ.

Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM

Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM is an official topic in Module 1 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the imperfections in crystal. 4 understanding material science engineering: necessity for material science and engineering- components of material science and engineering. classification of materials with applications- engineering requirement of materials - selection of materials. crystal structure: structure of atoms and molecules -crystal structure- crystal geometry, structure of solids, methods of determining structures-x-ray diffraction technique-nano crystalline solids. imperfection in crystals: types of imperfection. point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, sem, tem should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of

imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM
 definition/meaning
 steps, application
 Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the structural properties of crystals

Outline the importance of material science and

M1.01		2
Understanding	technology.	
M1.02	Identify the various materials and its applications	3
Understanding.		
M1.03	Outline crystal structure	4
Understanding.		
M1.04	Summarize the imperfections in crystal.	4
Understanding		

Contents:

Material science engineering: necessity for material science and engineering- components of material science and engineering.

Classification of materials with applications- engineering requirement of materials - selection of materials.

Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids.

Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Summarize the properties of materials

This module is presented from the official syllabus scope for Material Science and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the properties of materials
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties

Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties is an official topic in Module 2 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the mechanical properties of materials 5 understanding summarise the elastic and hardness properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Summarise the mechanical properties of materials
5 Understanding Summarise the elastic and hardness properties
definition/meaning .
steps, application . Technical terms English-
.

4 Understanding of material

4 Understanding of material is an official topic in Module 2 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of material means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding of material
 definition/meaning steps, application Technical terms English-

Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the

Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the is an official topic in Module 2 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the electrical properties of the material. 4 understanding summarise the magnetic properties of the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the
definition/meaning .
steps, application . Technical terms English-
.

3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity-superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior - ferro electricity - Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples.

3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity-superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior - ferro electricity - Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples. is an official topic in Module 2 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity-superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior - ferro electricity - Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding material properties of materials: mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; rockwell hardness, brinell hardness, vickers hardness. electrical properties: electrical conduction and conductivity- superconductivity- semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior – ferro electricity – piezo electricity. magnetic properties: magnetism- types of magnetism- dia-magnetism- para-magnetism- ferro-magnetism- anti-ferro-magnetism- ferri-magnetism-with examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity- superconductivity- semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior – ferro electricity – Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience,

toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness.
 Electrical properties: electrical conduction and conductivity-superconductivity-semi
 conductivity- electrical conduction in polymers and ionic ceramics- dielectric
 behavior – ferro electricity – Piezo electricity. Magnetic properties: magnetism- types
 of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-
 magnetism- Ferri-magnetism-with examples.
 definition/meaning
 steps, application
 Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
 Educational explanations below are separate elaboration.

Summarize the properties of materials		
M2.01	Summarise the mechanical properties of materials	5
Understanding	Summarise the elastic and hardness properties	
M2.02		4
Understanding	of material	
M2.03	Summarise the electrical properties of the material.	4
Understanding	Summarise the magnetic properties of the	
M2.04		3
Understanding	material	
	Series Test – I	1
Contents:		
Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior.		
Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness.		
Electrical properties: electrical conduction and conductivity-superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior –		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 3

Summarize the processing and application of metal alloys

This module is presented from the official syllabus scope for Material Science and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the processing and application of metal alloys
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal

Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal is an official topic in Module 3 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the processing of metal alloys 4 understanding summarise the thermal processing of metal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Summarise the processing of metal alloys 4
Understanding Summarise the thermal processing of metal
definition/meaning . . . , steps, application
 Technical terms English-

4 Understanding alloys

4 Understanding alloys is an official topic in Module 3 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding alloys using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding alloys should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding alloys means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding alloys ഓക്സീകരണത്തിന്റെ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ഓക്സീകരണത്തിന്റെ ഓക്സീകരണ, steps, application ഓക്സീകരണത്തിന്റെ ഓക്സീകരണം. Technical terms English-ഓക്സീകരണത്തിന്റെ നിർവ്വചനം.

Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of

Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of is an official topic in Module 3 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret corrosion in metal and alloys. 4 understanding summarize the properties and application of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇരുമ്പിന്റെ Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of ഇരുമ്പിന്റെ ഉപയോഗം definition/meaning ഇരുമ്പിന്റെ ഉപയോഗം. ഇരുമ്പിന്റെ ഉപയോഗം, steps, application ഇരുമ്പിന്റെ ഉപയോഗം ഇരുമ്പിന്റെ ഉപയോഗം. Technical terms English-ഇരുമ്പിന്റെ ഉപയോഗം.

5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries.

5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries. is an official topic in Module 3 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding metal and alloys processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. thermal processing- annealing-heat treatment of steel-precipitation hardening. corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. properties and application of metals and alloys: engineering materials - ferrous metals - iron and their alloys iron and steel - iron carbon equilibrium diagram. non-ferrous metals and alloys - aluminum, copper, zinc, lead, nickel and their alloys with reference to the application in chemical industries. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of

metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries. **മലയാളം** definition/meaning **മലയാളം**. **മലയാളം** **മലയാളം** **മലയാളം**, steps, application **മലയാളം** **മലയാളം** **മലയാളം**. Technical terms English-**മലയാളം** **മലയാളം**.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the processing and application of metal alloys

M3.01 Summarise the processing of metal alloys 4
Understanding

Summarise the thermal processing of metal

M3.02 4
Understanding

alloys

M3.03 Interpret corrosion in metal and alloys. 4
Understanding

Summarize the properties and application of

M3.04 5
Understanding

metal and alloys

Contents:

Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding.

Thermal processing- annealing-heat treatment of steel-precipitation hardening.

Corrosion of metals: electro chemical consideration-corrosion rates – prediction of

corrosion-forms of corrosion – corrosion prevention methods.

Properties and application of Metals and alloys: Engineering materials - ferrous metals -

Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals

and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 4

Summarize the characteristics and processing of inorganic materials

This module is presented from the official syllabus scope for Material Science and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the characteristics and processing of inorganic materials
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding ceramics Outline the characteristics and processing of

3 Understanding ceramics Outline the characteristics and processing of is an official topic in Module 4 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding ceramics Outline the characteristics and processing of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding ceramics outline the characteristics and processing of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding ceramics Outline the characteristics and processing of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-3 Understanding ceramics Outline the characteristics and processing of **കെറാമിക്സ്** **definition/meaning** **കെറാമിക്സ്**. **കെറാമിക്സ്** **കെറാമിക്സ്** **steps, application** **കെറാമിക്സ്** **കെറാമിക്സ്**. Technical terms English- **കെറാമിക്സ്** **കെറാമിക്സ്**.

3 Understanding polymers. Outline the characteristics and processing of

3 Understanding polymers. Outline the characteristics and processing of is an official topic in Module 4 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding polymers. Outline the characteristics and processing of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding polymers. outline the characteristics and processing of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding polymers. Outline the characteristics and processing of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-3 Understanding polymers. Outline the characteristics and processing of polymer definition/meaning polymer . polymer polymer polymer , steps, application polymer polymer . Technical terms English- polymer polymer .

3 Understanding composites. Identify the types and applications of inorganic

3 Understanding composites. Identify the types and applications of inorganic is an official topic in Module 4 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding composites. Identify the types and applications of inorganic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding composites. identify the types and applications of inorganic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding composites. Identify the types and applications of inorganic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-3 Understanding composites. Identify the types and applications of inorganic composites. definition/meaning of composites. steps, application of composites. Technical terms English- composites.

3 Understanding materials. Characteristic and processing of ceramics: ceramics - crystal structure - silicates and carbon- mechanical properties - ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories - polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 [https://nptel.ac.in/chemical engineering 2](https://nptel.ac.in/chemical%20engineering) [https://www.classcentral.com/course/edx-materials-science-and-engineering-8150 3](https://www.classcentral.com/course/edx-materials-science-and-engineering-8150) [https://www.edx.org/learn/materials-science 4](https://www.edx.org/learn/materials-science) <https://ocw.mit.edu/courses/materials-science-and-engineering/>

3 Understanding materials. Characteristic and processing of ceramics: ceramics - crystal structure - silicates and carbon- mechanical properties - ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories - polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 [https://nptel.ac.in/chemical engineering 2](https://nptel.ac.in/chemical%20engineering) [https://www.classcentral.com/course/edx-materials-science-and-engineering-8150 3](https://www.classcentral.com/course/edx-materials-science-and-engineering-8150) [https://www.edx.org/learn/materials-science 4](https://www.edx.org/learn/materials-science)

<https://ocw.mit.edu/courses/materials-science-and-engineering/> is an official topic in Module 4 of Material Science and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding materials. Characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon- mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories – polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 [https://nptel.ac.in/chemical engineering](https://nptel.ac.in/chemical%20engineering) 2 <https://www.classcentral.com/course/edx-materials-science-and-engineering-8150> 3 <https://www.edx.org/learn/materials-science> 4 <https://ocw.mit.edu/courses/materials-science-and-engineering/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding materials. characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon- mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation.

characteristic and processing of polymers: polymer classification- molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. types and application of inorganic materials: ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. ceramics, glass and refractories – polymer types and application- advanced materials (biomaterials, nanomaterials and composites) with special reference to the applications in chemical industries. text / reference t/r book title/author materials science and engineering an introduction; william d callister, david g t1 rethwisch r1 materials science and engineering- a first course; v. raghavan online resources sl.no website link 1 [https://nptel.ac.in/chemical engineering](https://nptel.ac.in/chemical%20engineering) 2 <https://www.classcentral.com/course/edx-materials-science-and-engineering-8150> 3 <https://www.edx.org/learn/materials-science> 4 <https://ocw.mit.edu/courses/materials-science-and-engineering/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding materials. Characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon- mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories – polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 https://nptel.ac.in/chemical engineering 2 https://www.classcentral.com/course/edx-materials-science-and-engineering-8150 3 https://www.edx.org/learn/materials-science 4 https://ocw.mit.edu/courses/materials-science-and-engineering/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3 Understanding materials. Characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon-mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite-composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories – polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 [https://nptel.ac.in/chemical engineering](https://nptel.ac.in/chemical%20engineering) 2 <https://www.classcentral.com/course/edx-materials-science-and-engineering-8150> 3 <https://www.edx.org/learn/materials-science> 4 <https://ocw.mit.edu/courses/materials-science-and-engineering/>
 പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms English- പദപരിചയം പദപരിചയം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Summarize the characteristics and processing of inorganic materials

Outline the characteristics and processing of

M4.01 3

Understanding

ceramics

Outline the characteristics and processing of

M4.02 3

Understanding

polymers.

Outline the characteristics and processing of

M4.03 3

Understanding

composites.

Identify the types and applications of inorganic

M4.04 3

Understanding

materials.

Series Test – II 1

Contents:

Characteristic and processing of ceramics: ceramics – crystal structure -
silicates and

carbon- mechanical properties – ceramic fabrication techniques - glass forming -
particulate

forming- cementation.

Characteristic and processing of polymers: polymer classification- Molecular
characteristics

(classification only) - polymerization- polymer additives- forming techniques for

Module study routine: Read the source wording, make a one-page concept map,
practise one short answer and one structured answer, then review the Malayalam
support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding technology.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding technology..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify the various materials and its applications 3 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Identify the various materials and its applications 3 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline crystal structure 4 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline crystal structure 4 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science and engineering-components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the imperfections in crystal. 4 Understanding Material science engineering: necessity for material science*

and engineering- components of material science and engineering. Classification of materials with applications- engineering requirement of materials - selection of materials. Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids. Imperfection in crystals: types of imperfection. Point imperfection and types- miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding of material. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding of material*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise the electrical properties of the material. 4 Understanding Summarise the magnetic properties of the.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity- superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior - ferro electricity - Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding material Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior. Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness. Electrical properties: electrical conduction and conductivity- superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior - ferro electricity - Piezo electricity. Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding alloys. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding alloys*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of. (3 marks)

Answer: Begin with the standard definition or identity of *Interpret corrosion in metal and alloys. 4 Understanding Summarize the properties and application of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types-casting operation and types- powder metallurgy-welding. Thermal processing-annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates - prediction of corrosion-forms of corrosion - corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries.. (3 marks)

Answer: Begin with the standard definition or identity of 5 *Understanding metal and alloys Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding. Thermal processing- annealing-heat treatment of steel-precipitation hardening. Corrosion of metals: electro chemical consideration-corrosion rates - prediction of corrosion-forms of corrosion - corrosion prevention methods. Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding ceramics Outline the characteristics and processing of. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding ceramics Outline the characteristics and processing of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding polymers. Outline the characteristics and processing of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding polymers*. Outline the characteristics and processing of. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding composites. Identify the types and applications of inorganic. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding composites*. Identify the types and applications of inorganic. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding materials. Characteristic and processing of ceramics: ceramics - crystal structure - silicates and carbon- mechanical properties - ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories - polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources SI.No Website Link 1

[https://nptel.ac.in/chemical engineering 2](https://nptel.ac.in/chemical%20engineering)

<https://www.classcentral.com/course/edx-materials-science-and-engineering-8150> 3 <https://www.edx.org/learn/materials-science> 4

<https://ocw.mit.edu/courses/materials-science-and-engineering/>. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding materials. Characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon- mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation. Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers. Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites. Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories – polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. Text / Reference T/R Book Title/Author Materials Science and Engineering an Introduction; William D Callister, David G T1 Rethwisch R1 Materials Science and Engineering- A First course; V. Raghavan Online Resources Sl.No Website Link 1 [https://nptel.ac.in/chemical engineering 2](https://nptel.ac.in/chemical%20engineering) <https://www.classcentral.com/course/edx-materials-science-and-engineering-8150> 3 <https://www.edx.org/learn/materials-science> 4 <https://ocw.mit.edu/courses/materials-science-and-engineering/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding technology.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Summarise the mechanical properties of materials 5 Understanding Summarise the elastic and hardness properties.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Summarise the processing of metal alloys 4 Understanding Summarise the thermal processing of metal.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding ceramics Outline the characteristics and processing of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link [https://nptel.ac.in/chemical engineering](https://nptel.ac.in/chemical%20engineering)
- <https://www.classcentral.com/course/edx-materials-science-and-engineering-8150>
<https://www.edx.org/learn/materials-science>
- <https://ocw.mit.edu/courses/materials-science-and-engineering/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5073C. Verify institutional updates against the current official curriculum.